

OsmoFlux — Equation Reference

1D Axial Series-Current RED Model · Activity-Corrected Nernst EMF · Dual-Mechanism Fouling · Rohan Agarwal

1. EMF & Open-Circuit Voltage

Nernst EMF — per axial segment j

$$\mathcal{E}_j = \alpha \cdot \frac{2RT}{F} \ln \left(\frac{\gamma_{H,j} C_{H,j}}{\gamma_{L,j} C_{L,j}} \right)$$

α = ALPHA_PERM = 0.975 (Fumasep FKS-PET-130)
 γ_H, γ_L = activity coefficients — Robinson & Stokes (1959), log-linear interp.
 $C_{H,j}, C_{L,j}$ = segment concentrations from C_H_IN_MOL_L, C_L_IN_MOL_L
 $j = 1 \dots N_SEG$ (default 40 segments per cell pair)

Stack open-circuit voltage [V_OC_EQ_V]

$$V_{OC} = \sum_{j=1}^{N \times N_{seg}} \mathcal{E}_j - N \cdot V_{elec}$$

N = N_CELL_PAIRS (swept 2–80 in geometry optimisation)
 V_{elec} = 0.1 V per electrode pair — Veerman (2010)

2. Internal Resistance & Power

Segment resistance (Thévenin equivalent)

$$R_j = \frac{ASR_{CEM} + ASR_{AEM}}{\beta_{mem} \cdot A_{mem}} + \frac{d_{sp}}{\beta_{sol} \cdot \sigma_j \cdot A_{mem}}$$

$ASR_{CEM_OHM_M2}$ = $3.6 \times 10^{-4} \Omega \cdot m^2$ (FKS-PET-130 midpoint)
 $ASR_{AEM_OHM_M2}$ = $2.35 \times 10^{-4} \Omega \cdot m^2$ (FAS-PET-130 midpoint)
 β_{mem} = 1.10, β_{sol} = 1.56 — spacer shadow factors (Vermaas 2014)
 σ_j = NaCl conductivity, log-log interp.; d_{sp} = GAP_M = $2 \times 10^{-4} m$

Stack internal resistance [R_INT_EQ_OHM]

$$R_{int} = \sum_{j=1}^{N \times N_{seg}} R_j$$

Series-current topology: one current I through all $N \times N_{seg}$ segments

Maximum power density [P_DENS_W_M2]

$$P_{dens} = \frac{V_{OC}^2}{4 R_{int} N A_{mem}}$$

Analytical result at matched load $R_{load} = R_{int}$
 P_TOTAL_W = $P_DENS_W_M2 \times N_CELL_PAIRS \times A_MEM_M2$

3. Fouling Mechanisms

CDF — divalent ion scaling (inlet-heavy)

$$\Delta R_{CDF,j} = \beta \cdot f_{div} \cdot \left(\frac{C_{H,j}}{C_{ref}} \right)^n \cdot ASR_{CEM}$$

β = FOULING_BETA $\in [0, 1]$; f_{div} = DIVALENT_FRACTION
 C_{ref} = FOULING_C_REF_MOL_L = 0.5 mol/L; n = FOULING_N_EXP = 0.5
Source: Długołęcki (2010); penalty peaks at $j=1$ (inlet) where C_H is highest

Exponential biofilm — outlet-heavy $\phi(j)$

$$\Delta R_{bio,j} = \beta \cdot ASR_{CEM} \cdot \frac{\exp \left(\frac{j-1}{N_{seg}-1} \right) - 1}{e-1}$$

$\phi(j=1) = 0$ (no biofilm at inlet); $\phi(j=N_{seg}) = \beta \cdot ASR_{CEM}$ (max at outlet)
Normalisation by $(e-1)$: boundary conditions satisfied exactly
Biofilm builds where ionic gradient is weakest (toward outlet)

Total segment resistance with fouling

$$R_j^{foul} = R_j + \Delta R_{CDF,j} + \Delta R_{bio,j}$$

Combined penalty 4–7 pp below independent sum (sub-additive)
Mechanism: both terms compete for the same Thévenin driving force

4. Cascade Architecture & Harm Reduction

Harm reduction index HR

$$HR = 1 - \frac{C_{H,out} - C_{ref,sea}}{C_{H,in} - C_{ref,sea}}$$

$C_{H,out}$ = brine salinity leaving RED system [mol/L]
 $C_{H,in}$ = C_H_IN_MOL_L (raw brine inlet)
 $C_{ref,sea}$ = ambient receiving-body salinity [mol/L]
 $HR = 1 \rightarrow$ full compliance; $HR = 0 \rightarrow$ no improvement

Salt extracted per cascade stage s

$$\dot{m}_s = I_s \cdot N_{cp,s} \cdot \frac{M_{NaCl}}{F}$$

$I_s = V_{OC,s} / (2R_{int,s})$ = optimal current [A]
 $N_{cp,s}$ = N_CELL_PAIRS in stage s ; M_{NaCl} = 58.44 g/mol

Cascade power gain G vs single stack

$$G = \frac{P_{cascade} - P_{single}}{P_{single}} \times 100\%$$

Positive only above $C_H_IN_MOL_L \gtrsim 1.5$ mol/L
Up to +418% at hypersaline ($C_H = 3.0$, $S = 10$, FOULING_BETA = 0.8)
Fouling shift: optimal N_CELL_PAIRS per stage falls as FOULING_BETA rises

5. Techno-Economic Analysis

Net present value (primary metric)

$$NPV = \sum_{t=1}^T \frac{R_t - C_t}{(1+r)^t} - C_{CAPEX}$$

R_t = electricity revenue + avoided brine disposal cost ($\sim \$0.04/m^3$)
Carlsbad P50 NPV = \$181.8 M; $P(NPV > 0) = 100\%$ (5,000-sample Monte Carlo)

LCOE (reference only — not primary metric)

$$LCOE = \frac{C_{CAPEX} + \sum C_t(1+r)^{-t}}{\sum E_t(1+r)^{-t}}$$

P50 = \$3.05/kWh — misleading without disposal credit
NPV is the correct metric; value driven by avoided disposal cost

6. 1D Series-Current Solver

Axial salt conservation (per segment j)

$$C_{H,j+1} = C_{H,j} - \frac{\dot{n}_j}{Q_H} \quad \dot{n}_j = \frac{I \cdot N_{cp}}{F \cdot N_{seg}}$$

I = single stack current shared by all $N_{cp} \times N_{seg}$ segments (series topology)
 $\dot{n}_j$ = molar salt transfer rate at segment j [mol/s]
 Q_H = volumetric brine flow rate (Q_H_M3_S) [m³/s]

Max-power condition (bisection root)

$$f(I) = E_{total}(I) - 2 I R_{total}(I) = 0$$

$E_{total}(I) = \sum \mathcal{E}_j(C_{H,j}(I), C_{L,j}(I))$ — depends on I via salt depletion
 $R_{total}(I) = \sum R_j^{foul}(C_{H,j}(I))$ — depends on I via local concentrations
Solved by bisection over $I \in [0, I_{max}]$; N_SEG = 40 axial segments

Optimal current and power at solution

$$I^* = \frac{E_{total}(I^*)}{2 R_{total}(I^*)} \quad P_{max} = \frac{E_{total}^2}{4 R_{total}}$$

Non-trivial: E_{total} and R_{total} both vary with I (salt depletion couples them)
Reduces to $P = V_{OC}^2 / (4 R_{int})$ only when concentrations are I -independent